



## Question One

a. i). This is a cross-sectional study. If we assume that each subject has independent outcomes, then  $\text{Var}(\bar{Y}^A) = \text{Var}(\bar{Y}^B) = \frac{2\sigma^2}{k}$

and  $\bar{Y}^A \perp \bar{Y}^B$ . Thus,  $\text{Var}(\hat{\tau}_a) = \text{Var}(\bar{Y}^A) + \text{Var}(\bar{Y}^B) = 4\sigma^2/k$

ii). For a single subject  $i$ , define  $S_i = Y_{i1} - Y_{i0}$ .  $\text{Var}(S_i) = \text{Var}(Y_{i1} - Y_{i0}) = \text{Var}(Y_{i1}) + \text{Var}(Y_{i0}) - 2\text{Cov}(Y_{i1}, Y_{i0}) = 2\sigma^2 - 2\rho\sigma^2 = 2\sigma^2(1-\rho)$

Since each subject is independent with each other, then we have  $\text{Var}(S_k) = \text{Var}(S_1)$ ,  $k=1, 2, \dots, k$

Therefore,  $\text{Var}(\bar{\Delta}^A) = \text{Var}(\bar{\Delta}^B) = \frac{2\sigma^2(1-\rho)}{k/2} = \frac{4\sigma^2(1-\rho)}{k}$ ,  $\text{Var}(\hat{\tau}_b) = 8\sigma^2(1-\rho)/k$

iii). Define  $S_k = Y_{k1} - Y_{k2}$ ,  $\text{Var}(S_k) = 2\sigma^2(1-\rho)$ .

$\hat{\tau}_c = \bar{Y}_1^A - \bar{Y}_2^B = \frac{1}{k} \sum_{k=1}^k (Y_{k1} - Y_{k2})$ ,  $\text{Var}(\hat{\tau}_c) = 2\sigma^2(1-\rho)/k$

iv). Define  $\mu_k^A = \frac{Y_{k1} + Y_{k2}}{2}$ , and  $\bar{Y}^A = \frac{2}{k} \sum_{k=1}^k \mu_k^A$ . Similar to treatment B,  $\bar{Y}^B = \frac{2}{k} \sum_{k=1}^k \mu_k^B$ .  $\hat{\tau}_d = \bar{Y}^A - \bar{Y}^B$

$\text{Var}(\mu_k^A) = \frac{1}{4} [\text{Var}(Y_{k1}) + \text{Var}(Y_{k2}) + 2\text{Cov}(Y_{k1}, Y_{k2})] = \sigma^2(1+\rho)/2$

$\text{Var}(\bar{Y}^A) = \text{Var}(\bar{Y}^B) = \frac{2\sigma^2(1+\rho)/2}{k/2} = \sigma^2(1+\rho)/k$

$\text{Var}(\hat{\tau}_d) = 2\sigma^2(1+\rho)/k$

b.  $300000/500 = 600$  times of measurements.

For methods (i), (ii), (iii) and (iv).

|   | i   | ii  | iii | iv  |
|---|-----|-----|-----|-----|
| K | 600 | 300 | 300 | 300 |

Var  $\sigma^2/150$   $2\sigma^2(1-\rho)/15$   $\sigma^2(1-\rho)/30$   $\sigma^2(1+\rho)/30$

Given different values of  $\rho = \{0.2, 0.5, 0.8\}$ , the variances of four methods are shown below:

| $\rho$ | i              | ii                                | iii                               | iv                                |
|--------|----------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 0.2    | $\sigma^2/150$ | $32 \times \frac{\sigma^2}{150}$  | $0.8 \times \frac{\sigma^2}{150}$ | $1.2 \times \frac{\sigma^2}{150}$ |
| 0.5    | $\sigma^2/150$ | $2 \times \frac{\sigma^2}{150}$   | $0.5 \times \frac{\sigma^2}{150}$ | $1.5 \times \frac{\sigma^2}{150}$ |
| 0.8    | $\sigma^2/150$ | $0.8 \times \frac{\sigma^2}{150}$ | $0.2 \times \frac{\sigma^2}{150}$ | $1.8 \times \frac{\sigma^2}{150}$ |

If we only compare the variance, then we should choose method (iii).

c. For a single subject with  $m$  times of measurements, the cost is  $250(H+m)$ . The number of recruitments are shown below:

|                   | i              | ii                     | iii                    | iv                     |
|-------------------|----------------|------------------------|------------------------|------------------------|
| # of measurements | 1              | 2                      | 2                      | 2                      |
| Costs / person    | 500            | 750                    | 750                    | 750                    |
| K                 | 600            | 400                    | 400                    | 400                    |
| Var               | $\sigma^2/150$ | $\sigma^2(1-\rho)/150$ | $\sigma^2(1-\rho)/200$ | $\sigma^2(1+\rho)/200$ |

Given  $\rho = \{0.2, 0.5, 0.8\}$ , the variances are:

| $\rho$ | i                               | ii                               | iii                               | iv                                |
|--------|---------------------------------|----------------------------------|-----------------------------------|-----------------------------------|
| 0.2    | $4 \times \frac{\sigma^2}{600}$ | $96 \times \frac{\sigma^2}{600}$ | $2.4 \times \frac{\sigma^2}{600}$ | $3.6 \times \frac{\sigma^2}{600}$ |

|     |                                 |                                   |                                   |                                   |
|-----|---------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 0.5 | $4 \times \frac{\sigma^2}{100}$ | $6 \times \frac{\sigma^2}{100}$   | $15 \times \frac{\sigma^2}{100}$  | $4.5 \times \frac{\sigma^2}{100}$ |
| 0.8 | $4 \times \frac{\sigma^2}{100}$ | $2.4 \times \frac{\sigma^2}{100}$ | $0.6 \times \frac{\sigma^2}{100}$ | $5.4 \times \frac{\sigma^2}{100}$ |

Therefore, we should still choose method iii based on variance.

Question 2.